YEAR 9 PHYSICAL SCIENCES
INDICATORS CHECK LIST

NAME:

INDICATOR	I CAN DO THIS
1. Explain how static electricity is produced. 2. State and apply the charge law. 3. Describe the following examples of static electricity in the real world: lightning, brushing hair, carpet, surface of cars	
4. Identify materials as conductors or insulators. 5. Define current electricity as the flow of electrons. 6. Draw the symbols for the main components (resistor, battery, globe, ammeter, voltmeter and switch) of an electric circuit. 7. Draw simple series and parallel circuit diagrams.	
8. Define current and state that the unit is the ampere (A). 9. Define potential difference and state that the unit is the Volt (V). 10. Define resistance and state that the unit is the Ohm (Ω) 11. Describe the energy transformation of a globe in a circuit 12. Discuss the advantages and disadvantages of series and parallel connections. 13. State Ohm's law (V = IR) 14. Interpret graphical representation of Ohm's Law. 15. Use V = IR to perform calculations. 16. Describe how an electromagnet is produced and its applications in the real world.	
23. Define light as a form of energy. 24. State the law of reflection on a plane mirror. 25. Draw a labelled ray diagram to show light reflection in a plane mirror (no image location). 26. Draw a labelled ray diagram to show the effects of concave and convex mirrors on parallel rays of light (no image location). 27. Identify the following uses of concave mirrors: car headlights, flashlights 28. Identify the wide of field of view of convex mirrors and its significance for the following uses: security mirrors in shops and driveways, mirrors in cars.	
29. Describe refraction as the bending of light rays as they move through media of different density. 30. Draw a labelled ray diagram to show the effects of concave and convex lenses on parallel rays of light. 31. Identify the following uses of convex lenses: magnifying glass, telescopes, microscopes, human eyes 32. Label the lens, retina and cornea on a diagram of the mammalian eye. 33. Describe the functions of the lens, retina and cornea. 34. Discuss the eye defects - Long sightedness and short sightedness. 35. Explain how spectacles can be used to fix long sightedness and short sightedness 36. List the components of the electromagnetic spectrum as radio waves, microwaves, infrared, visible light, ultraviolet, x-rays and gamma rays. 37. Define a wave as a method of transferring energy. 38. Classify water waves as longitudinal (uses a medium to travel) and light waves as transverse (travels with or without medium). 39. Label the crest, trough, frequency, wave length, amplitude and a complete cycle on a wave diagram. 40. State that Hertz is the unit for frequency and is defined as the number of wavelengths per second. 41. Use values from a wave diagram to calculate frequency.	